

EMPLOYEE MANAGEMENT SYSTEM

A Java Servlet & JSP Based Web Application

Submitted by: Vimalraj Vijayakandipen

Technology Stack: Java, JSP, Servlets, Tomcat, PostgreSQL

Overview

This project is a Java Web Application developed using Servlets, JSP, and Tomcat. It demonstrates Servlet lifecycle, ServletContext, ServletConfig, HttpSession, Cookies, RequestDispatcher, sendRedirect, and JSP integration for building a simple Employee Management System.

Problem Statement

Develop a Java Web Application to manage employee records using JSP and Servlets. The application must implement CRUD operations and demonstrate Servlet concepts like Context, Config, Session, and Cookies.

Application Modules

1. Home Page (index.jsp) – Defined as welcome file with navigation links. 2. Add Employee – JSP form to input details. Uses default department from ServletConfig. 3. View Employees – Displays all employees with Edit/Delete links. 4. Edit Employee – Loads existing data, updates map, stores session & cookie. 5. Delete Employee – Removes employee and redirects to list.

Core Features

- Use of ServletContext for global data storage
- Use of ServletConfig for default parameters
- Use of HttpSession and Cookies for state management
- Forward and Redirect handling
- JSP for presentation layer

Technologies Used

- Java 17
- JSP & Servlet (Jakarta EE)
- Apache Tomcat 10+
- PostgreSQL / H2 Database
- HTML, CSS
- JDBC

Servlet Concepts Demonstrated

ServletContext – Stores all employees globally
ServletConfig – Provides default department
HttpSession – Stores last updated Employee ID
Cookies – Stores last updated

Employee name RequestDispatcher – Forwards edit data to JSP sendRedirect – Used after create/update/delete Welcome File – index.jsp Load-on-startup – Initializes servlet on app startup

Testing Steps

1. Deploy project in Apache Tomcat.
2. Access <http://localhost:8080/EmployeeManagementSystem/>
3. Add, Edit, and Delete employees.
4. Validate last updated details from Session and Cookies.

Conclusion

The project successfully demonstrates Servlet and JSP working together in an MVC structure, showing session tracking, cookies, servlet lifecycle, and context configuration management.